

MACHINE LEARNING FUNDAMENTALS SERIES

**Machine Learning Fundamentals:
Bias and Variance***Technical Documentation on Model Flexibility, Overfitting, and Trade-off Optimization*

Source: StatQuest with Josh Starmer — Machine Learning Fundamentals

1 Introduction and Problem Formulation

When developing predictive machine learning models, a primary goal is to approximate the true underlying relationship between input features and target variables based on observed data [1].

Consider the clinical/biological scenario presented in the source video: predicting a **mouse's height** based on its **weight** [1].

- **Observed Pattern:** Light mice tend to be short, and heavier mice tend to be taller. However, beyond a certain weight threshold, mice cease getting taller and instead become obese, causing the height curve to plateau [1].
- **Mathematical Challenge:** The exact mathematical equation describing this biological relationship is typically unknown in real-world applications [1]. Machine learning algorithms are used to approximate this hidden relationship [1].

To evaluate model performance objectively without self-deception, the available dataset is split into two disjoint subsets [2]:

1. **Training Set (Blue Data Points):** Used exclusively to fit model parameters [2].
2. **Testing Set (Green Data Points):** Held out during training and used solely to evaluate how well the trained model generalizes to new, unseen data [2].

2 Core Terminology and Concepts

2.1 Bias (Underfitting)

Definition: Bias refers to the inability of a machine learning algorithm to capture the true underlying structure or relationship in the data [2].

- **Example Algorithm: Linear Regression** (Least Squares) [2].
- **Behavior:** Linear regression fits a rigid *straight line* to the training data [2]. Because a straight line lacks the mathematical flexibility to curve, it can never replicate the true curved relationship (arc) between weight and height [2].
- **Characteristics:** High bias leads to **underfitting**—the model is too simple to perform well even on the training set [2].

2.2 Variance (Overfitting)

Definition: Variance refers to the variability of model predictions across different datasets (specifically, the difference in fit/performance between the training set and the testing set) [4].

- **Example Algorithm: High-Degree Polynomial Regression** (Squiggly Line) [2, 3].
- **Behavior:** The high-degree polynomial fits a highly flexible *squiggly line* that hugs every individual point in the training set [2, 3].
- **Characteristics:** High variance leads to **overfitting**—the model learns the random noise/fluctuations of the training set rather than just the general trend [4, 5]. It performs exceptionally well on training data but fails drastically on testing data [3, 4].

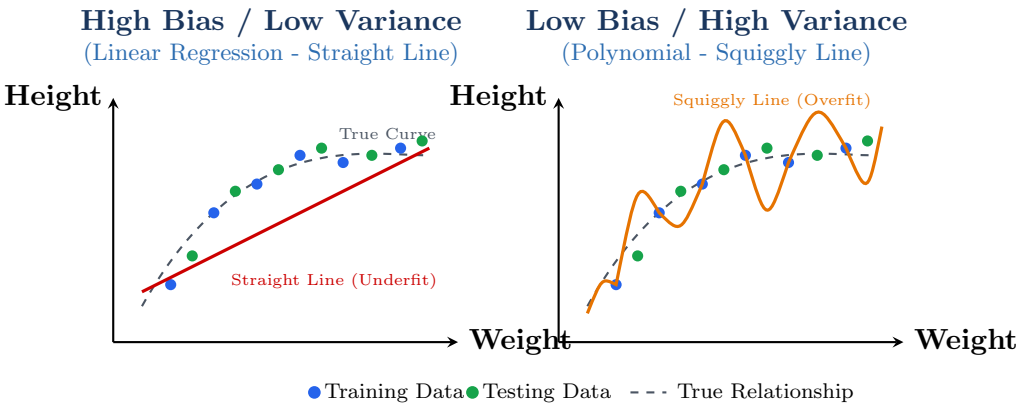


Figure 1: Visual Comparison of Bias and Variance in Fitting Mouse Weight vs. Height Data [1–4].

3 Quantitative Evaluation via Sum of Squares (SS)

To measure model fit mathematically, we compute the **Sum of Squared Residuals (SS)** [3]:

$$\text{Sum of Squares (SS)} = \sum_{i=1}^N (y_i - \hat{y}_i)^2 \tag{1}$$

where y_i represents the actual mouse height and $\hat{y}_i$ represents the model’s predicted height [3]. The distances are squared so that positive and negative errors do not cancel each other out [3].

Table 1: Comparison of Errors (Sum of Squares) Across Training and Testing Sets [3, 4].

Model Type	Training (SS _{Train})	Error	Testing (SS _{Test})	Error	Generalization Status
Straight Line (Linear)	Moderate / High <i>(Line cannot curve)</i> [2, 3]	<i>(Consistent with training error)</i> [4]	Moderate	<i>(Consistent with training error)</i> [4]	Underfit (High Bias, Low Variance) [4, 5]
Squiggly Line (Polynomial)	≈ 0 <i>(Passes through all blue points)</i> [3]	Very High <i>(Severe errors on green points)</i> [3, 4]	Very High	<i>(Severe errors on green points)</i> [3, 4]	Overfit (Low Bias, High Variance) [4, 5]

Key Insight: In the competition on the *training set*, the squiggly line wins [3]. However, in the true competition on the *testing set*, the straight line wins [3, 4]. The squiggly line demonstrates **high variability** because its performance degrades dramatically when evaluated on new data [4].

4 The Bias-Variance Tradeoff

4.1 Finding the "Sweet Spot"

In machine learning, the ideal algorithm satisfies two conditions simultaneously [5]:

1. **Low Bias:** Sufficient flexibility to accurately model the true relationship [5].
2. **Low Variance:** Produces consistent, reliable predictions across different datasets [5].

Total expected prediction error can be decomposed into three mathematical components:

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error} \tag{2}$$

As model complexity increases, bias decreases monotonically while variance increases monotonically [5]. Finding the **optimal model complexity ("Sweet Spot")** balances these two opposing forces to minimize total test error [5].

4.2 Algorithmic Methods to Achieve the Sweet Spot

To discover and enforce this sweet spot in practical machine learning workflows, three fundamental advanced techniques are commonly employed [5]:

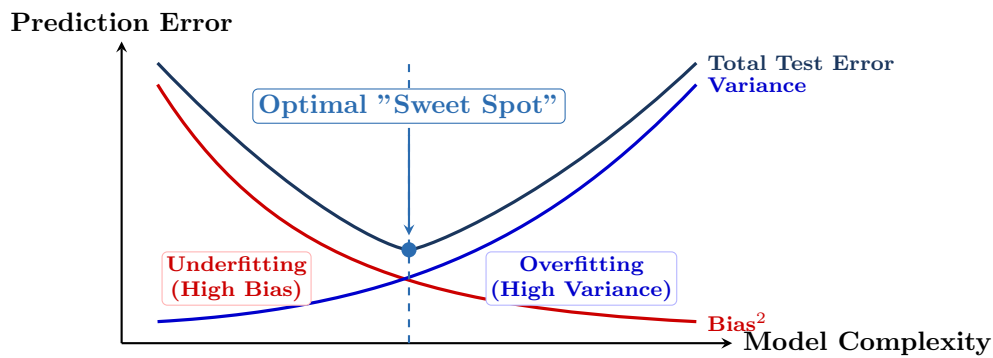


Figure 2: The Bias-Variance Tradeoff Curve illustrating Underfitting, Overfitting, and the Optimal Sweet Spot [5].

1. **Regularization (e.g., Ridge / Lasso Regression):** Penalizes model complexity to prevent over-flexible parameter estimates [5].
2. **Bagging (Bootstrap Aggregating, e.g., Random Forests):** Reduces variance by combining predictions from multiple decision trees trained on bootstrap samples [5].
3. **Boosting (e.g., AdaBoost, Gradient Boosting):** Reduces bias sequentially by building ensemble models that focus on misclassified samples [5].

5 Summary of Key Terminology

Bias: The systematic error introduced by approximating a complex real-world relationship with a model that is too simple [2].

Variance: The sensitivity of a model to small fluctuations in the training dataset [4].

Underfitting: Occurs when a model has high bias and fails to fit either the training or testing data adequately [2, 5].

Overfitting: Occurs when a model has high variance, fitting training data almost perfectly but failing on test data [4, 5].

Sweet Spot: The balanced level of model complexity where total test error is minimized [5].